

MD RAIHAN CHOWDHURY

Software & AI Systems Engineer

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Links: [Portfolio](#) | [GitHub](#) | [LinkedIn](#)



CAREER OBJECTIVE

To secure a challenging position where I can apply my expertise in full-stack web development, AI/ML integration, and system architecture. A recent B.Sc. CSE graduate from North South University with hands-on experience building production applications, winning competitive datathons, and contributing to AI-powered product development. I aim to grow as an engineer while delivering meaningful software solutions.

WORK EXPERIENCE

- Triliant Data**
Software Engineer

Dhaka, Bangladesh
February 2026 – Present

 - Engineering end-to-end AI workflows and REST API integrations for client products, reducing system response times and improving automated data processing reliability.
 - Architected and deployed **Annotigo**, an automated AI data-annotation platform; streamlined dataset labeling flows using custom API webhooks and background processing tasks.
- Synnax Lab**
Data Scientist

Remote
November 2024 – July 2026

 - Built and deployed AI-driven predictive models for the Synnax Predictive Credit Intelligence Protocol.
 - Engineered scalable data pipelines to process high-volume financial datasets with daily model retraining.
- Freelance**
Web Developer

Remote / Local
2025 – 2026

 - mjapparelsLtd.com:** Designed and developed a full corporate website for a garments company.
 - hopetkf.org.au:** Built a non-profit organization website for an Australian charity foundation.

FEATURED PROJECTS

- Agentic PR Fixer — Autonomous AI Code-Review & Self-Healing Agent**
FastAPI, LangGraph, Docker SDK, Redis, PyGithub, Python 3.11

[\[GitHub\]](#)

 - Engineered a LangGraph and Redis state machine to automate GitHub PR reviews and stateful patch generation.
 - Built a self-healing loop parsing pytest errors alongside sandboxed Docker SDK execution and HITL review gates.
- MinutelyAI — AI Meeting Intelligence Application**
Next.js 16, FastAPI, LangGraph, Whisper, Tailwind CSS

[\[Live Demo\]](#) · [\[GitHub\]](#)

 - Built an agentic AI app converting audio and text into structured meeting summaries, action items, and decisions.
 - Orchestrated a 4-node LangGraph pipeline integrating local OpenAI Whisper and DeepSeek V3 for low-latency analysis.
- TaskSync-HQ — Multi-Tenant Real-Time Kanban Engine**
Next.js 15, TypeScript, Supabase, PostgreSQL, Tailwind CSS, Vitest

[\[Live Demo\]](#) · [\[GitHub\]](#)

 - Built a multi-tenant Kanban platform with real-time Supabase CDC sync, optimistic UI, and 24 Vitest tests.
 - Designed a relational PostgreSQL schema using Supabase RLS policies for multi-tenant RBAC enforcement.

EDUCATIONAL QUALIFICATIONS

Examination	Institution	Result	Year
B.Sc. in CSE	North South University, Dhaka	CGPA: 3.23 / 4.00	2020 - 2025
A Levels	British Council, Bangladesh	GPA: 2.5	2018 - 2019
O Levels	Maple Leaf International School	GPA: 5.0	2017 - 2018

RESEARCH

Illegal Garbage Detection Using Computer Vision

Journal Paper — In Progress

- Trained YOLOv8, YOLOv11, Detectron2, and Mask R-CNN on a custom dataset; achieved 96% mAP (IoU 0.5).
- Built complete detection pipeline integrated with a mobile app, cloud backend, and IoT hardware.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript (ES6+), Java, SQL, HTML5, CSS3, Fortran
Frontend:	React.js, React Native, React Query, Next.js, Tailwind CSS
Backend:	FastAPI, Flask, Node.js, Express, LangGraph, Redis, RESTful API
ML / AI:	PyTorch, scikit-learn, YOLOv8/v11, Mask R-CNN, LLM integration, Prompt Engineering, AI-Assisted Development (Claude Code, OpenAI Codex, Cursor)
Data & Databases:	Pandas, Numpy, Data Pipelines, SQL Databases, Financial Dataset Processing
DevOps & Tools:	Git, GitHub, Docker SDK, Vercel, Vitest, Linux, API & Webhook Integrations, Automated Workflows, System Monitoring & Debugging , Raspberry Pi, IoT systems

LANGUAGE PROFICIENCY

Bangla	Native
English	Fluent

ACHIEVEMENTS, AWARDS, & CERTIFICATIONS

Honors & Competition Awards

- 2nd Place — HerWILL AI for Equality Datathon
- Top 20 — Synnax Datathon
- Top 50 — Robi Datathon 3.0 Qualifier
- Shortlisted — Cisco IoT Hackathon 2024
- Award Recipient — NSU ACM SC Innovation Challenge 2024 (AI and IoT)

Professional Certification

- Foundations of Data Science — Coursera (Grade: 95.50%) [[View Credentials](#)]
- Foundations: Data, Data, Everywhere — Coursera (Grade: 98.25%) [[View Credentials](#)]

EXTRACURRICULAR ACTIVITY

Member — North South University Communications Club (NSUCC)

Jan 2021 – Feb 2022

- Attended workshops and seminars to develop communication and public speaking skills; supported overall event execution.